METRIC

MIL-H-83282C
INTERIM AMENDMENT 1
31 JANUARY 1997

MILITARY SPECIFICATION

HYDRAULIC FLUID, FIRE RESISTANT, SYNTHETIC HYDROCARBON BASE, METRIC, NATO CODE NUMBER H-537

This interim amendment is approved for use within the Naval Air Systems Command, Department of the Navy, with MIL-H-83282C, dated 25 March 1986, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Bottom of page: DELETE the 'Beneficial comments' block and SUBSTITUTE the following:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100 B120-3, Highway 547, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

2.1.1: DELETE "FED-STD-313 - Material Safety Data Sheets, Preparation and Submission of" and SUBSTITUTE "FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities."

2.1.1: DELETE "MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes."

2.1.1, just above 2.1.2, ADD "(Unless otherwise indicated, copies of specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)"

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:45Z
Check the source to verify that this is the current version before use.

2.2, ADD:

"ASTM-D4898 - Insoluble Contamination of Hydraulic Fluids by Gravimetric Analysis, Standard Test Method for

ASTM-D5185 - Determination of Additive Elements, Wear Metals, and Contaminants in Used Lubricating Oils by Inductively Coupled Plasma Atomic Emission Spectrometry, Standard Test Method for."

2.2, DELETE the street address for the American Society for Testing and Materials and SUBSTITUTE "100 Barr Harbor Drive, West Conshohocken, PA. 19428-2959."

2.2, ADD:

"AMERICAN SOCIETY FOR QUALITY CONTROL

ASQC Z1.4- Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted)

(Applications for copies should be addressed to American Society for Quality Control, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)"

4.3.1, line 16 and 17: DELETE "Aircraft and Crew Systems Technology Directorate, Code 60612, Naval Air Development Center, Warminster, PA 18974," and SUBSTITUTE "Commander, Attn: Code 4.3.4.1 Building 2188, Naval Air Warfare Center Aircraft Division, 22347 Cedar Point Road, Unit 6, Patuxent River MD 20670-1161"

4.3.1: DELETE and SUBSTITUTE:

"4.3.1 Qualification sample. The qualification test sample shall consist of two 1-gallon containers of the finished fluid, two quarts of the base stock without additives, two quarts of the blending fluids used, eight ounces of the anti-wear additive, one ounce of the oxidation inhibitor and one ounce each of any additional additives used in the finished fluid. In the event that the additives are supplied as concentrated solutions, an equivalent quantity of the solution shall be furnished. Upon receipt of authorization (see 6.3) from theCommander, Attn: Code 4.3.4.1 Building 2188, Naval Air Warfare Center - Aircraft Division, 22347 Cedar Point Road, Unit 6, Patuxent River MD 20670-1161 to submit samples for qualification testing, samples shall be forwarded as directed. The samples shall be accompanied by a test report from the manufacturer or a commercial laboratory containing complete information as to the source and type of base stock and additive materials used, the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:45Z
Check the source to verify that this is the current version before use.

formulation and composition of the finished fluid, and laboratory data showing quantitative results of all the tests required by this specification, except storage stability. Separate qualification inspections shall be required for each base stock used. A material safety data sheet shall accompany the samples being submitted for test. Samples shall be identified as follows:

- Sample for qualification inspection
- HYDRAULIC FLUID, FIRE RESISTANT, SYNTHETIC HYDROCARBON BASE, METRIC, NATO CODE NUMBER H-537
- Name of ingredient (for ingredient material)
- Name of manufacturer
- Product code number
- Date of manufacture
- Submitted by (name) and (date) for qualification inspection in accordance with the requirements of MIL-H-83282C under authorization of (reference authorizing letter) (see 6.3)."

PAGE 8

4.4, line 8, 4.4.4, line 3, and 4.4.5, line 4: DELETE "MIL-STD-105" and SUBSTITUTE "ASQC Z1.4."

PAGE 14

6.3, line 10: DELETE "AIR-5304C2, Department of Navy, Washington, DC 20361," and SUBSTITUTE "AIR-4.3.4, Arlington, VA 22243."

6.3, lines 13 and 14: DELETE "Aircraft and Crew Systems Technology Directorate, Code 60612, Naval Air Development Center, Warminster, PA 18974," and SUBSTITUTE "Commander, Attn: Code 4.3.4.1 Building 2188, Naval Air Warfare Center - Aircraft Division, 22347 Cedar Point Road, Unit 6, Patuxent River MD 20670-1161."

6.4: DELETE and SUBSTITUTE "6.4 Standard rubber samples. Samples of the standard synthetic rubber NBR-L (see 3.3.4) for the test specified in table VIII may be obtained from Wynn's Precision Company, 708 West 22nd Street, Tempe, AZ 85282."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:45Z
Check the source to verify that this is the current version before use.

PAGE 16

TABLE II, ADD the following entry:

Characteristic	Requirement
Barium content, parts per million (ppm) (mass percent), maximum	10

PAGE 18

TABLE VI, ADD the following entry:

Inspection	Paragraph	
	Requirement	Test method
Barium content	3.3	4.5.2

PAGE 19

TABLE VIII, DELETE entry for "Solid particle contamination" and SUBSTITUTE the following:

Characteristic	Test method	
	FED-STD-791	ASTM
Solid particle contamination	---	D4898 2/
Barium Content	---	D5185

PAGE 20

TABLE VIII, footnote 2/: DELETE and SUBSTITUTE: "2/ For ASTM-D4898, a single 0.45 micron, cellulose membrane filter shall be used. Particle contamination may also be measured, following FED-STD-791, Method 3012, using an automatic particle counter capable of measuring the limits defined in table III; the HIAC/ROYCO brand, model 8011, has been found capable of measuring these limits."

PAGE 26

DELETE DD Form 1426, "Standardization Document Improvement Proposal," and SUBSTITUTE the form on the last page of this amendment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:45Z
Check the source to verify that this is the current version before use.

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy - AS
(Project No. 9150-N830)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:45Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. **The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.**

2. **The submitter of this form must complete blocks 4, 5, 6, and 7.**

3. **The preparing activity must provide a reply within 30 days from receipt of the form.**
   NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-H-83282C	2. DOCUMENT DATE (*YYMMDD*) 860325

3. DOCUMENT TITLE

HYDRAULIC FLUID, FIRE RESISTANT, SYNTHETIC HYDROCARBON BASE, AIRCRAFT, METRIC, NATO CODE NUMBER H-537

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial: (2) DSN: (*If Applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE NUMBER (*Include Area Code*) (1) Commercial: (908) 323-7488 (2) DSN: 624-7488
c. ADDRESS (*Include Zip Code*) CODE 414100 B120-3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office, 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:45Z
Check the source to verify that this is the current version before use.

COMMANDER
NAVAL AIR WARFARE CENTER
AIRCRAFT DIVISION
CODE 414100B120-3
HIGHWAY 547
LAKEHURST, NJ 08733-5100

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:45Z
Check the source to verify that this is the current version before use.